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Human Rationality

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Keywords

judgment, decision-making, reasoning, argument, individual rationality, collective rationality

Abstract

This article provides a critical overview of research on human rationality. Rationality research poses a number of unusual challenges to psychologists. For one, it is unusually interdisciplinary and involves research conducted in adjacent disciplines (e.g., economics, education, communication, computer science and philosophy), not all of which are accessible with psychological training. What underlies this diversity, however, is an arguably even more unusual feature: the fact that even purely descriptive research, focused on what we actually do, cannot proceed without reference to normative considerations, that is, considerations of what we ought to do. Empirical results can thus be understood only with some understanding of the relevant norms of rationality. This article introduces the range of relevant frameworks, followed by examples of the different ways these frameworks are put to use. The bulk of the article then surveys research findings on human rationality across the core areas of (probability) judgment, reasoning, decision-making, and argumentation. Two final sections provide cross-cutting themes, one on the contrast (and interrelationship) between individual and collective rationality and one on the unique challenges of linking rationality research to real-world concerns.

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1. INTRODUCTION

Questions about human rationality have an odd place within psychology. On the one hand, they are wrapped up with the very notion of what it means to be human: "ratio" and "rational" in their original sense of reason and with a capacity for reason have been used to differentiate humans from nonhuman species for thousands of years (see, e.g., Anton 1996; Knauff & Spohn 2021, chapter 1). On the other hand, research on human rationality is an anomaly and outlier within psychology because it is inherently concerned not only with how humans are, descriptively, but also with how that compares to how they, in some sense, ought to be. Research on rationality intrinsically involves comparing human behavior to a standard or norm and asking to what extent, and by what means, human behavior manages to agree with that norm.

This has meant that there is less focus on psychological process in rationality research than is common in other areas of psychology. It has also meant a strong interdisciplinary character. A close overlap with adjacent fields (biology, pharmacology, sociology, or neuroscience, to name but a few) is not an unusual feature of psychology, but the disciplines that overlap in the case of rationality research—primarily philosophy, economics, statistics, and computer science—are different from those with overlap to other areas of psychology. The extent of human rationality (and how rationality might be defined) is a topic of interdisciplinary concern; furthermore, rational

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actor models feature in many disciplines as a methodological tool—in psychology and cognitive science, but also in sociology, neuroscience, and economics. These further interdisciplinary links have made other disciplines not just consumers of psychological research on rationality but also producers of parallel programs of descriptive enquiry of their own.

The goal of this article is to provide an overview of research on human rationality that clarifies these multiple threads. To this end, it focuses very much on questions: What normative standards exist for assessing rationality? How are these used in research? Who conducts this research and why? What open questions exist and what future developments might we foresee? The article does not seek to provide a comprehensive summary of research on rationality per se, for reasons that will become apparent.

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2. PART I: NORMS OF RATIONALITY

2.1. Logic

The longest-standing, and most fundamental, standard of rationality is classical logic (for an excellent introduction, see Gamut 1991). Classical logic illuminates the notions of necessarily true and necessarily false. With that, it provides the notion of logical consequence or deductive validity, whereby the conclusion of an argument follows from one or more premises by necessity, such that one cannot, without self-contradiction, accept the truth of the premises but reject the truth of the consequence. Early psychological research on logical reasoning undermined the long-held view that the laws of logic are the laws of thought (see Henle 1962; for an overview, see Oaksford & Chater 2007). The subsequent development of reasoning research paralleled a shift seen across a range of disciplines, from philosophical work on argumentation [Toulmin 2003 (1958)] to the development of artificial intelligence (Rumelhart et al. 1986), that came to recognize the limits of logic in everyday argument and reasoning (Oaksford & Chater 1991). While the notions of logical necessity and logical contradiction provide fundamental constraints on the quality of human argument and reasoning, most of everyday argument and reasoning goes beyond deduction, and with that, it goes beyond what can be captured with classical logic. Outside of mathematics, most issues of human concern involve statements that are only contingently true or false, not true or false by logical necessity. There have been many attempts to expand the reach of logic through the development of so-called nonclassical logics (Prakken & Vreeswijk 2002), which are equipped for dealing with the uncertainty and non-monotonicity that are an inherent part of reasoning or for arguing about the kinds of contingent claims humans are most concerned with.[1] Non-classical logics, however, have garnered little attention in psychological research (for an exception, see, e.g., Stenning & van Lambalgen 2008).

There are likely multiple reasons behind this: a gap between disciplines, but maybe also the fact that there are many approaches and rival nonclassical systems, as well as the fact that their normative foundations are typically not well developed. Finally, probability theory is a prominent rival with respect to handling uncertainty and non-monotonicity.

2.2. Probability

Probability theory expands the notion of contradiction that is central to classical logic to the notion of probabilistic coherence. For example, it is not probabilistically coherent to believe that there is a 70% chance of an event (e.g., rain) occurring and a 40% chance of it not occurring (i.e.,

[1]Non-monotonicity refers to the possibility that additional information may overturn an already established conclusion.

not rain) or to believe that it is more likely that Liverpool will beat Manchester United and lose against Newcastle than it is likely that Liverpool will beat Manchester United (the so-called conjunction fallacy; see Tversky & Kahneman 1983; specifically for betting versions, see Sides et al. 2002, Nilsson & Andersson 2010). From the axioms of probability theory follows Bayes' theorem. This is applied by Bayesians also as a normative procedure for (diachronic) belief revision, that is, the determination of a new degree of belief in light of evidence (Lin 2024). This commitment to belief revision via Bayes' rule (also known as conditionalization), together with the notion that degrees of belief should be assigned in line with the probability calculus (also known as probabilism), constitutes Bayesianism as, arguably, the most widespread normative standard for rationality research within psychology. It encompasses also the use of Bayesian Belief Networks as tools for representing multivariable problems to simplify Bayesian computations (Pearl 1988).

Normative uses of probability theory are not limited to Bayesianism, however. Studies of contingency judgments, for example, have compared human performance to ΔP [$\Delta P = P(A|B) - P(A|notB)$], which itself may be related to statistical measures such as Chi square (Ward & Jenkins 1965), and the classical statistical inference methods analysis of variance (ANOVA) has doubled as a (seemingly) normative model for how lay reasoners should approach causal attribution problems (Kelley 1973, Försterling 1989; but see, e.g., Newcombe & Rutter 1982).

2.3. Decision Theory

Closely related to probability theory is decision theory, which sets out how we should combine rational beliefs with rational preferences in order to choose between uncertain outcomes. Decision theory sets out a minimal account of rationality for such choices that is independent of the specific content of the options under consideration. Rather, it rests only on the desirability the different possible outcomes hold for us and on their probabilities of occurring (Steele & Stefánsson 2020). Specifically, canonical decision theory—that is, expected utility theory (EUT)—states that we should prefer the option with the greatest expected (subjective) value: In deciding which of multiple options to select, we should weight each option's subjective value (or utility) by its probability of occurrence. This is well illustrated by the kinds of monetary lotteries or gambles that historically constituted the fruit fly of decision-making research: Would you, for example, prefer option A, a 50% chance of winning 100 pounds, or option B, a 40% chance of winning 130 pounds?

Decision theory renders this a comparison between two expected utilities: 0.5×100 and 0.4×130, that is, 50 versus 52 (on the assumption that the subjective value of 100 and 130 pounds to you corresponds straightforwardly to their numeric value). This simple calculation shows option B to be the higher value, and hence preferrable, option.

Crucially, there is nothing in decision theory that commits you to equating the monetary value with your own subjective value (in fact, applications of EUT standardly assume that you do not, and that money has diminishing marginal utility, that is, the more money you have, the less valuable a given additional amount becomes). Nor is the applicability of EUT restricted to explicitly numeric outcomes. Rather, it can be shown that as long as your preferences respect a set of very general constraints (more on this below), it follows that they can be represented by a cardinal utility function, that is, by a function that captures not only whether an option is more valued than another but also how much more it is valued. This, in principle, gives decision theory an incredibly broad potential domain of application.

2.4. Game Theory

Despite its broad scope, there are decisions that are not covered by decision theory. Chief among these are strategic choices, that is, choices where my best course of action depends on others'

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choices, and vice versa. Such interacting decisions are the focus of game theory (Davis 2012). Game theory presupposes individual rationality in the decision-theoretic sense in order to provide answers and insight into best responses in strategic contexts.

In order to provide normative guidance, it characterizes strategic choices in terms of the respective payoffs that each agent (player) would receive under the different combinations of choices possible. This manner of representing strategic contexts supports analytic possibilities such as the identification of choice combinations where neither player can unilaterally improve their outcome by switching to a different choice—the so-called Nash equilibrium (on its uses, see Holt & Roth 2004). It also affords the identification of different classes of games (Colman 2016), such as zero-sum games and coordination games (Cooper & Weber 2020). It also captures particular individual games, such as the Prisoner's Dilemma (Dawes 1980) or the Ultimatum Game (Güth et al. 1982), which, though typically conceived with a specific narrative, may manifest in many different contexts involving different payoffs, as it is the structural relationships between different payoffs that ultimately matter, not their specific values.

In this, game theory shares the characteristics of all the frameworks discussed: The rules they all posit are abstract inasmuch as a task's specific content and context determine the key quantities on which they rest (e.g., truth values for statements in the case of logic, subjective degrees of belief for Bayesian probability, utilities for EUT, payoffs in games), but beyond that, those specific contents and contexts do not matter. Moreover, once those quantities are fixed, prescriptions may follow algorithmically—though not all questions of interest may be computable (see, e.g., Boolos et al. 2002), and those that are may not necessarily be answerable with practically feasible computational resources of space or time (Dagum & Luby 1993, van Rooij et al. 2018).

2.5. Reasonableness

Despite the broad scope of these frameworks, there remain open questions about the relationship between these standards of rationality and wider notions of reasonableness. On the one hand, the rationality norms discussed might be too strict in that (at least in certain contexts) they set out idealized requirements that are beyond the means of actual human agents. This may be seen to undercut their normative force by virtue of the principle, invoked in ethics and law, that humans cannot be bound to do the impossible (the *impossibilium nulla obligatio est* of Roman law; see Corner & Hahn 2013). At the same time, though, this impossibility need not undermine their merit as targets for us to approximate any more than my inability to calculate in my head the square root of 12,526 makes it irrelevant that there is, in fact, a correct answer. Not only are human capabilities not fixed (e.g., I can calculate quantities with a calculator that I could never calculate in my head), but also the targets for approximation can provide valuable analytical tools, as will be discussed below.

The question, however, remains of how humans should proceed where these rationality standards are impossible or too onerous in practice. Just because the ideal answer cannot be reached doesn't mean that all others are equally good or bad. Bounded rationality (Simon 1957) concerns itself with useful accommodations or alternatives when human resource limitations are taken into account. For example, there may be heuristics—simple rules of thumb—that work as well (or even better) than resource-bound optimal procedures given the right conditions or environment (see, e.g., Gigerenzer et al. 1999, Juslin et al. 2009 for specific examples).

At the same time, one might consider the rational norms offered by the preceding frameworks to be too limited. Specific content and context may give rise to intuitively relevant considerations that go beyond what is captured by these frameworks. Good decisions, for example, may in some cases require wisdom or judgment (in a wider sense) to be good. Making such concerns more precise is aided by closer consideration of the ongoing development of the relevant normative frameworks, which we discuss next.

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3. DEVELOPMENT OF NORMATIVE SYSTEMS

The just described normative frameworks are well established, provide widely (but not universally) accepted standards of what it might mean to be rational, and form the backbone of psychological research on human rationality. Nevertheless, the development of formal frameworks seeking to define rational behavior is very much an ongoing project. In particular, there is active development on the normative foundations of these frameworks, that is, on the potential justifications from which they might derive their normative status. There is also research that identifies important gaps, that is, aspects of thought and action that intuitively feel like they should be subject to normative prescriptions but currently fall outside of coverage of these systems.

For example, there are long-standing projects to extend classical logic to cover inferences involving possibility (Garson 2024) or obligation that have not resolved into a single accepted system (Harman 1972, Burgess 1999). In the context of probability theory, there is research on identifying more general rules for belief revision that encompass conditionalization (revision by Bayes' rule) but also extend to updating beliefs in light of new conditional probabilities (e.g., Douven 2012, Eva & Hartmann 2018, Eva et al. 2020, Hartmann et al. 2020, Strößner & Hahn 2025) or to including new events or possibilities to be considered (e.g., Williamson 2003, Fuchs et al. 2024). There is also a strong sense that there should be meaningful things to say about more radical forms of uncertainty (Knight 1921) for which the probability calculus seems insufficient [see e.g., the special issue introduced by Romeijn & Roy (2014)], for example, uncertainty that does not even let us conceptualize the space of possible outcomes.

These issues are inherited by decision-making, but decision-making also conflicts with intuition on further issues. One such issue is the putative rationality of deliberate ignorance (Hertwig & Engel 2021): Can it be rational to reject new information—for example, the outcome of a genetic test for a severe, untreatable, future medical condition? The standard decision-theoretic answer is no (see Hertwig & Engel 2021, chapter 14). Another issue is whether decision theory deals appropriately with variance (i.e., risk; see, e.g., Lopes 1982, Tversky & Bar-Hillel 1983), a matter that continues both to drive debate around the normative foundations of decision theory and to prompt the development of alternative systems (for a general review of alternative decision theories, see, e.g., Sugden 2004).

4. NORMATIVE FOUNDATIONS

4.1. Logic and Probability

Asking about the normative foundations of theories of rationality is to ask from what they derive their prescriptive value. Where does the "ought" come from? Why and in what sense are the prescriptions of these frameworks rational? Merely asserting that something is normative is as insufficient for rationality as it is elsewhere.

This question has received less attention for classical logic than it has for probability theory, let alone decision theory or game theory (cf. Steinberger 2022), presumably in part because the core defect identified by logic seems obviously bad: Logical contradictions are statements that are necessarily false. Furthermore, given that logical inconsistency also gives rise to a lack of probabilistic coherence, one can appeal to some of the foundational arguments for probability theory as well.

The best-known of these is the so-called Dutch book argument. A Dutch book is a bet one is guaranteed to lose, whatever happens, and incoherent beliefs make one vulnerable to Dutch books. A sprawling literature has discussed this argument (for extensive reviews see, e.g., Hajek 2008) and how far it actually goes.

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More recently, research has sought to base the normative justification for Bayesian probability on its connection with the accuracy of our beliefs (e.g., Rosenkrantz 1992, Joyce 1998, Pettigrew 2016). Accuracy-based justifications involve the use of a scoring rule to measure the accuracy of probabilistic forecasts, that is, to determine how good people are at making probability judgements about future events (e.g., predicting the weather; Winkler & Murphy 1968). A scoring rule assigns credit for correct predictions and penalties for incorrect predictions. The rule then combines these into a total score that represents overall accuracy. Rosenkrantz (1992) shows that updating by Bayes' rule maximizes the expected score after sampling; this means other updating rules will be less efficient in the sense that they will require larger samples, on average, to be as accurate. This holds for any way of measuring accuracy that involves a so-called proper scoring rule, that is, a scoring rule that will yield highest scores when agents report honestly their actual degrees of belief (e.g., there is no incentive for agents to hedge their bets). Furthermore, this optimality of Bayesian conditionalization with respect to maximizing accuracy not only holds for disinterested inquiry but also holds when actions dependent on our beliefs about the world are at stake: Using Bayesian conditionalization to update our beliefs upon having sampled evidence maximizes expected utility (Brown 1976, Rosenkrantz 1992). Finally, Bayesianism (as the assignment of probabilities in accordance with the probability axioms and updating via Bayes' rule) follows from the simple premise that an agent ought to approximate the truth and hence seek to minimize inaccuracy if inaccuracy is measured using the Brier score (Brier 1950): Being Bayesian will minimize the inaccuracy of the agent's beliefs across all possible worlds the agent is conceptually able to distinguish and hence, in principle, to entertain (Pettigrew 2016).[2]

The use of scoring rules links also to a further line of foundational argument for probability theory: its inevitability. Lindley (1982) shows that under extremely general assumptions about how to score the accuracy of an estimate of uncertainty, the estimate is, in effect, a probability. Snow (1995, 1998), relatedly, shows that probability as a representation of beliefs about uncertainty follows from a small number of simple assumptions, with the consequence that supposedly alternative schemes for representing uncertainty turn out to be probability in disguise (Heckerman 1986).

4.2 Decision Theory

With respect to the normative foundations of decision theory, the easiest way to see its normative appeal is to consider contexts in which we are inclined to think in terms of repeated outcomes. The expectation or expected value, here, is just the average in the sense of the mean. Maximizing expected value, consequently, simply means choosing in such a way as to maximize getting those things we value (and minimize those that we do not) in the long-run average.

This perspective is useful, also, because many of the concerns surrounding the normative status of decision theory focus on cases in which considering only the mean outcome—for example, to the exclusion of outcome variability—feels intuitively inadequate. One example of this is the fact that most would reject paying more than somewhere between 2 and 4 dollars to play a run of a game where a coin is flipped repeatedly and one wins 2^n dollars, where n is the number of coin tosses before the first heads occurs (i.e., Bernoulli's St. Petersburg Paradox; see Lopes 1982). That price corresponds closely to the median outcome of that game but is far from the expected value, which is driven by the astronomical value of extremely long runs of heads that are possible but equally astronomically improbable.[3]

[2]This holds with the restriction that these possible worlds are finite, a restriction that seems fine for creatures with finite resources and life spans.

[3]In fact, the expected value is technically infinite.

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The more familiar foundation of decision theory, though, is its axiomatic derivation. Here it is shown that from a few basic assumptions (axioms), in particular constraints on preferences, it follows that decisions in accordance with those axioms can be represented as maximizing expected utility (von Neumann & Morgenstern 1947). For example, preferences must be transitive (if one prefers A to B and B to C, one ought also prefer A to C), a constraint that seems intuitively appealing in as much as violating it allows one to be turned into a money pump—that is, a sequence of swaps of items A, B, and C in exchange for money leaves one with the very set of items one started with, but poorer for it (see Tversky 1969, Bar-Hillel & Margalit 1988, Birnbaum & Gutierrez 2007)—though it may prove difficult to robustly demonstrate such violations in practice (Regenwetter et al. 2011). These axioms feature prominently in psychological research on decision-making because patterns of choice that are qualitatively incompatible with these axioms allow one to establish irrationality without requiring access to either an agent's subjective probabilities or their subjective utilities (see, famously, Allais 1953, Kahneman & Tversky 1979). Despite this, however, serious debate about the adequacy of these axioms continues (see Anand 1987, 1995), as do attempts to develop alternatives to decision theory.

Overall, the normative status of decision theory seems more unsettled than that of probability theory, let alone classical logic. This holds even more so for game theory. Not only does game theory, once again, inherit the rationality assumptions of these other frameworks, but it may also seem less than obvious why the central concept of Nash equilibrium should coincide with a rational selection of strategy. One construal is in terms of advice (see Holt & Roth 2004): If one imagines oneself giving advice on the best strategy to each player individually, any advice that is not a Nash equilibrium implies that one is giving bad advice to at least one of the players. In other words, it would be better for some player to behave other than advised if all other players actually followed the advice they received. The crucial bit in this is "to each player individually," inasmuch as better options may be available if players act in concert. While the individualistic perspective may fit well with evolutionary contexts (Maynard Smith 1976), the game theory literature is awash with studies demonstrating that people choose with greater regard to cooperation than analyses centered on Nash equilibria would mandate (Kollock 1998).

This does not obviate the value of game theory as an analytic tool with which to identify relevant structural configurations governing social situations, but it is less obvious that one should equate rationality with an individualistic perspective when better outcomes are obtainable by joint action.

Furthermore, the already mentioned issue of idealization has a different resonance in the context of game theory. It is one thing to specify a rational ideal for a belief or decision of mine that I cannot actually attain in practice. This ideal may nevertheless serve as a meaningful goal. It is a wholly different matter to deem rational a response of mine that rests on assumptions about others that will not, or cannot, ever be accurate in practice. In keeping with this, game theory has sought (under the header of epistemic game theory; see, e.g., Perea 2014) to develop extensions that incorporate beliefs about other agents and beliefs about beliefs.

5. REASONABLENESS RECONSIDERED

The first thing that emerges from consideration of the respective normative foundations of the central frameworks for rationality is that the rationality they define is instrumental. They specify means to particular ends—in effect, tools. If I want to avoid Dutch books, efficiently use my evidence, avoid money pumps, or, more generally, maximize (in the long-run average) getting what I want, then they tell me what I ought to do. They do not, however, tell me that I must pursue these ends in the first place.

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Furthermore, other additional considerations may, at least in some circumstances, be of relevance to me that are not accommodated by these frameworks in their current form, and research on the normative foundations of these frameworks, in fact, helps clarify precisely what some of these may be. At the same time, the above discussion has highlighted some of the gaps in these frameworks—questions that feel like they should have at least partial normative answers but currently do not, and that reveal them to be ongoing projects.

There is thus more than enough in these frameworks of rationality to resist the idea that they exhaust our notion of reasonableness. To push back on overclaiming of their reach, one need not invoke additional, external considerations such as unease about the fact that some of the technical developments of these frameworks were tied up with US Cold War efforts (Erickson et al. 2013) or have been viewed as expressions of patriarchy (for a review, see Lloyd 2017). It is, however, worth briefly considering at least some of these criticisms here. One such line of criticism sees problems in the opposition between rationality and emotion, given the way this contrast has been mapped onto gender (see, e.g., England 1989).

The normative frameworks of rationality just outlined are, in fact, entirely silent on emotion. In that sense, they are neither "for" nor "against" it. Furthermore, they specify structural relationships (e.g., this consequence follows logically, these beliefs are coherent, this choice has higher expected utility, this combination of choices represents a Nash equilibrium) that seem useful. They do not specify processes. Decision theory, for example, provides an (instrumental) recommendation to choose the higher expected utility option. It does not tell me how I come to do that, and many different processes (and many more approximations) may lead to such a choice. Whether or not emotion is at odds with rationality (or whether that question even makes sense in that generality) depends on the extent to which emotion does or does not lead to outcomes at odds with these rational norms. There is an extensive literature on rationality and emotion (for a review see, Pham 2007). That literature has often had as its intuitive point of departure that the two are in conflict, but this stance is by no means universal, and research has equally sought to highlight the fact that emotion may be an important guide in these normative contexts (see, e.g., Bechara et al. 2000, Mellers et al. 2002).

At the same time, it seems worth stressing that the instrumental relationships set out in foundational research are statements of fact. My beliefs and choices are subject to the instrumental constraints that our frameworks of rationality outline, regardless of how I feel about these frameworks or the notion of rationality and reasonableness they embody.

In keeping with both of these points, feminist critiques of rationality have been controversial within feminist philosophy (see Alcoff 1995, Lloyd 2017). In particular, it has been pointed out that there are minimal standards of rationality that intellectual debate must presuppose in order to proceed at all, and which consequently cannot really be at issue. At the same time, that there are many grounds to not equate those standards of rationality with (all of) reason, such as the fact that reason might additionally include moral judgement (Nagl-Docekal 1999, Neiman 2009), echoes the verdict that already follows from closer consideration of extant normative frameworks of rationality themselves: As we saw above, they do not capture all aspects of reason.

6. PART II: THE MULTIPLE USES OF RATIONALITY NORMS IN EMPIRICAL RESEARCH

The notion of our core frameworks of rationality as tools is reflected also in their application in research. Within psychological research this encompasses, broadly, three uses: as standards for assessing competence, as targets for amelioration, and as methodological tools. These three uses will be discussed in turn.

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6.1. Competence

Large swathes of research on human rationality have ostensibly focused simply on trying to gauge levels of human competence. In other words, the question at issue is a question about what humans are like. Such rationality-focused research is not fundamentally different from psychological research aimed at understanding the capacity of short-term memory, reading ability, or the ease with which different types of category structures are learned. The motivation and ultimate goal of such research is to understand the nature of the cognitive system and the processes and mechanisms by which we solve tasks such as remembering, reading, or categorization. The only difference is that the performance standard against which competence is evaluated is itself sufficiently complex as to be the focus of sustained theoretical inquiry and development.

One difference that nevertheless emerges from that comparison is the level of scrutiny and contentiousness that accompanies claims about competence across these different areas. There have been many studies probing short-term memory span, including studies questioning the utility of the very concept. This does not, however, feel comparable in either volume or intensity to what has been referred to as the rationality wars (e.g., Gigerenzer 2025).

One difference is that the charge of irrationality goes to the heart of our self-perception as human beings in ways that limitations of short-term memory or difficulties in learning nonlinearly separable categories simply do not. The second is that these standards of rationality figure as a key building block in models and theories central to economics (and also parts of sociology). Consequently, there is a considerable investment in the question of the extent to which they might provide reasonable descriptive approximations to human behavior.

6.2. Amelioration

At the same time, research on rationality is conducted in the service of amelioration: finding ways to improve human performance. There exist extensive literatures, across multiple disciplines, that seek to teach people how to foster better argumentation and improve critical thinking (e.g., Kuhn & Crowell 2011), avoid logical fallacies and improve logical reasoning (e.g., Nunes et al. 2007), improve probabilistic reasoning and avoid common probability errors, improve decision-making, and develop techniques for avoiding or reducing bias (Larrick 2004, Soll et al. 2015). Arguably, one limitation of the sprawling interdisciplinary research literature concerned with rationality is the often somewhat limited integration between research concerned with determining competence and research concerned with amelioration. This seems to be to the detriment of both. Research on amelioration sometimes seems to lag behind the state of the art in research on competence with respect to putative failings. For example, it is relatively easy to find amelioration work that takes a theoretical stance on so-called logical fallacies (in particular, affirming the consequent; Oaksford & Chater 1994) that is at odds with research in the psychology of reasoning, which has developed a much more nuanced perspective over the last decades on when these inferences are and are not fallacious (more on this below). Conversely, by virtue of poor integration with research on amelioration, research concerned with gauging competence is missing out on useful information about boundary conditions.

6.3. Methodological Tools

Finally, considerations of rationality provide an important (and controversial) methodological tool. Multiple interrelated approaches exist that draw on optimal models in trying to understand human behavior.

6.3.1. Marr's levels of explanation. The first of these is Marr's classic articulation of what it means to understand a computational system (Marr 1982). According to Marr, a computational

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system must be understood on three levels: the computational level, the algorithmic level, and the implementation level. This is illustrated with the simple example of a cash register: A full understanding of a cash register involves understanding what it computes and why (computational level), how it algorithmically achieves those computations (algorithmic level), and how, finally, they are implemented in an actual physical device (implementation level). The computational level supplies a functional analysis whereby the cash register "does addition" because the social norms governing the purchase of multiple items stipulate that we owe the sum of the individual items (special offers aside). The algorithmic level identifies a particular procedure (for example, carry over the least significant digit), and the implementation level details how all of this is implemented with transistors and wires. All three levels are required because they constrain one another only loosely: Many different algorithms for addition exist, and a particular algorithm, in turn, can be given very different physical implementations. Crucially, the computational level asks questions not only about the "what" of the cash register but also about the "why":

> This whole argument is what I call the *computational theory* of the cash register. Its important features are (1) that it contains separate arguments about what is computed and why and (2) that the resulting operation is defined uniquely by the constraints it has to satisfy. (Marr 1982, p. 23)

Consequently, the computational-level theory appeals to factors beyond the device itself—in this case, the social conventions governing purchase. This emphasis on why opens the door for normative frameworks of rationality, because—by virtue of the instrumental value that underwrites their prescriptive force—they are inherently linked with questions about why. And, given the generality of these frameworks, those instrumental values are often of relevance when considering human behavior.

6.3.2. Rational analysis. This becomes even clearer with the methodological program of rational analysis (Anderson 1991, Chater & Oaksford 1999). Rational analysis seeks (*a*) a specification of the goals of the system and (*b*) a specification of the nature of the environment, and then, given both, it seeks (*c*) a derivation of an optimal behavior function. In the final step, actual behavior is explained as an approximation of that optimal function, if possible.

This approach, together with the popularity of computational-level theories, has considerably expanded the use of normative frameworks within psychology, because rational analysis, as a method and style of explanation, can be applied to behaviors, such as categorization (Anderson 1991), that are not necessarily construed as examples of inference, judgment, or decision-making (see Hahn 2014 for the wider implications of this point).

6.3.3. Ideal observer analysis. Similar to rational analysis, ideal observer analysis, as successfully used in the study of perception (e.g., Geisler 1987), uses the formal tools of probability and decision theory to specify a model of optimal performance. The focal constraint, though, is the available input for a given task, not a characterization of the goals of the system or the nature of the environment. Nor is the task complete once an approximate correspondence between ideal agent and actual human performance is established. Instead, human performance and ideal observer are brought into closer and closer correspondence in a process of iterative refinement that, bit by bit, incorporates capacity limitations of the human system into the ideal observer. This approach provides a tool for the elucidation of mechanism and process embedded in an overall account that seeks to understand the system as doing the best it can do given the available hardware.

6.3.4. Bounded optimality. While ideal observer analysis involves understanding the impact of constraints on performance, Howes et al.'s (2009) notion of cognitively bounded optimality conversely appeals to optimal models to help identify system constraints in the first place. One of

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the most fundamental challenges that behavioral sciences face is the incredible flexibility of human behavior. This can make it difficult to distinguish whether an observed pattern of performance reflects a particular mechanism or simply a reasonable response to the task that is selected among many precisely because it is reasonable (for a further example of this problem, see Chater 2009). Or, to rephrase this, any search for cognitive universals faces a credit assignment problem that needs to work out whether aspects of behavior that appear universal reflect hard constraints within the system or reflect consistent selection of an obvious best strategy. According to Howes et al. (2009), seeking to understand performance relative to optimal strategies (that are computationally explicit, with well-defined expected payoffs) may help solve this credit assignment problem.[4]

6.3.5. Task analysis. As should be apparent from the preceding sections, a significant part of the methodological use of normative theories in understanding behavior lies in their use as tools to elucidate mechanism. The normative theory effectively guides the systematic exploration of mechanism by providing an organizing framework to direct that search and constrain the search space in meaningful ways.

From that perspective, it might be tempting to view the normative aspect of the framework as an almost gratuitous, nonessential add-on, such that nothing is lost if one were, for example, to consider Bayesian reasoning as a merely descriptive computational-level theory—as seems to be advocated, for example, by Elqayam & Evans (2011), or, most recently, Crupi & Calzavarini (2023). Setting aside for the moment the tension with Marr's emphasis on why as part of a computational-level theory, and the implications of this for explanation, such a purely descriptivist perspective suffers another limitation. Specifically, it fails to appreciate the special status of rationality as a methodological tool for task analysis. Here, the emphasis is not on understanding the nature of the cognitive agent but rather on understanding more clearly the nature of the problem faced by that agent.

Two examples illustrate this further. The first of these stems from research on optimism bias as a seemingly pervasive phenomenon with considerable practical implications. According to this phenomenon, which has been the subject of well over a thousand studies, humans are unduly optimistic with respect to their chances of experiencing adverse life events. The notion of bias here refers to a systematic deviation from the true value of an estimate; as such, it indicates merely that people are systematically wrong, not that they are irrational (see Hahn & Harris 2014). The phenomenon, however, has widely been taken to be evidence of motivated reasoning whereby the desirability of a conclusion unduly shapes (distorts) reasoning such that the reasoner arrives at palatable results (Kunda 1990). Motivated reasoning itself is widely seen to be irrational, in keeping with the fact that normative theories for reasoning—in particular, logic and probability theory—seemingly have no place for the desirability of outcomes (but see also Hahn & Lewandowsky 2024).

Unfortunately, the overwhelming number of studies establishing a seemingly robust basis for optimism bias draws on a single method (Weinstein 1980). Harris & Hahn (2011) showed how this method gives rise to biased responses even for unbiased, wholly accurate Bayesian agents. This shows the method to be unsuitable and not fit for purpose. Moreover, the extent to which hypothetical data derived from such Bayesian agents fit known moderators of the optimism bias suggests that results may simply represent a statistical artifact and provide no insight into actual human behavior. A similar fate befell a newer literature on optimistic belief updating within cognitive neuroscience: Here, Shah et al. (2016) show how the core phenomenon of interest—that

[4]Confusingly, the term "bounded optimality" has also come to denote the project of resource-rational analysis (e.g., Lieder & Griffiths 2020), which is, in spirit, closer to ideal observer analysis.

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agents seemingly revise their beliefs more in response to good news than to bad news—once again arises with rational Bayesian agents given the nature of Bayesian belief revision and the probability scale.

These examples illustrate how seemingly intuitive experimental tasks and methods may be shown to be flawed through consideration of the performance of an optimal agent. Going one step further, finally, task analysis with normative theories can shape and alter our sense of what there is to be explained. The second example seeks to illustrate this. A topic that has attracted increasing interest in recent decades is testimony: Much of what we believe to know, we know through the testimony of others, and, intuitively, how much one adjusts one's beliefs in response to someone saying "It is snowing" should depend on how reliable/accurate that person is (Hahn 2024).

In many real-world contexts, however, this problem of determining the reliability of others' testimony is exacerbated by the fact that we are typically part of wider social networks of communicating agents. This means not only that the reliabilities we are trying to estimate are constantly changing as information is exchanged but also that an optimal agent would need to be able to factor in the wider network structure. This is because we need to avoid double-counting, that is, overweighting the same evidence that reaches us via different routes: One should (rationally) increase one's belief that there is a bear in the neighborhood more on receipt of reports of three independent sightings than when one receives reports from three different people that are ultimately all based on the same fourth source. In other words, to properly weight evidence, we need to factor in evidential dependence. Attempts to build rational models of agents communicating across social networks, however, reveal that this dependence problem cannot actually be solved in realistic contexts: Not only do we typically not know the structure of the entire social network beyond those we communicate with directly, but there is also no known normative framework that could adequately capture readily occurring network configurations (see Pilditch et al. 2025). This implies that estimating the reliability of our testimonial sources is, in a deep way, an unanswerable question. However, that means also that there is, in some ways, often no fact to the matter of how much the testimony of others should change our beliefs.

The fact that one cannot, for common real-world circumstances, define an optimal agent who prescribes how we should revise our beliefs in light of the testimony of others indicates that the processes of belief formation we all engage in are profoundly more challenging than one would intuitively assume. This has a number of other consequences for contemporary debate on issues such as misinformation in online social media. For one, the preceding task analysis suggests that we likely overestimate how much maintaining accurate beliefs in response to testimony is in our own power as opposed to a lucky outcome of inhabiting a benign information environment. In this latter example, the fact that a (fully) Bayesian, rational approach is not possible (even in theory, let alone in practice) reveals insight into the nature of the problem. That, in turn, constrains the directions that plausible investigation and plausible explanations might take.

6.3.6. Summary. In conclusion, there are important applications of the normative systems that seek to define human rationality that do not have, as their goal, to provide an answer to the question of how rational we are. Though these applications may shed light on that question as a by-product, their goal is not to establish the cognitive system as optimal (see also Griffiths et al. 2012 for related points on Bayesian modelling). Rather, interest lies in the consequences of proceeding from the assumption of a rational actor. Even where we come to understand the system as doing the best it can given a particular set of constraints, the point is not that the system is optimal, but rather that we have an explanation of how and why the system works that encompasses an understanding of both mechanism (what) and functional explanation (why).

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6.4. Open Questions

The methodological uses of considerations of rationality are worth emphasizing not only because they represent a significant proportion of research concerned with rationality but also because they inform current controversies and open debate.

There is ongoing controversy about the value of normative theories in the behavioral sciences (e.g., Elqayam & Evans 2011, Crupi & Calzavarini 2023), and a significant proportion of that controversy concerns itself with the proper relationship between the normative and the descriptive. One cannot, in general, conclude simply from the fact that something ought to be the case that it also actually is (and vice versa). The mere existence of a speed limit does not mean that is how fast I am driving; and, by the same token, whatever speed I am driving is not simply the speed I ought to be driving. In keeping with this, projects characterized by their authors as rational analysis, such as Oaksford & Chater's (1994) research on logical reasoning, have repeatedly been accused of involving fallacious inferences from "is" to "ought" (Elqayam & Evans 2011, Crupi & Calzavarini 2023). Importantly, however, the lack of a necessary connection between "is" and "ought" does not mean there is no connection (see also Spohn 2020, Corner & Hahn 2013). The speed limit may well constrain my driving speed (because I am trying to avoid a ticket), and, conversely, my actual driving speed may be informative with respect to how fast I should be driving, that is, whether I am currently in a 30-km/h, a 50-km/h, or an 80-km/h speed limit zone.

Evaluating the charge of fallacious is/ought inference consequently requires paying close attention to the precise role the normative theory is playing. There is a world of difference between (fallaciously) arguing that the speed limit is 30 km/h by virtue solely of me driving that speed and using my driving speed as an inferential cue. My choice of a driving speed is (sadly) not how speed limits are actually set where I live. My driving speed does not make the speed limit the way it is. It is the decision of a legally authorized body that sets the speed limit, not my driving. There is nothing wrong, however, with using my driving speed as evidence for the fact that the speed limit on this stretch of road is 30 km/h and not 80, given the additional assumption that I am, more likely than not, trying to drive at the right speed. And there is nothing wrong with the explanation that I am currently driving at 35 km/h, because the speed limit is 30 km/h and I am trying to approximate that. Finally, there is nothing wrong if the authorities tasked with setting the speed limit pick a speed limit on the basis of descriptive facts—say, that drivers driving at 30 km/h in a particular zone have considerably fewer accidents.

This is why it is so important to their application that our normative theories have an independent, normative foundation (as set out above). The moment one is appealing to more than the mere fact of my driving behavior as the thing that makes the norm a norm, there is no fallacy here.

These considerations also clarify what is problematic about the claim that we could see the same, or greater, methodological gains by just ignoring the normative aspect of, say, the Bayesian framework and treating it as a purely descriptive, computational-level theory. Doing so would mean losing the "why" that is an integral part of the computational-level theory: The fact that the normative theory can provide a functional explanation rests on its normative status. Likewise, it is the (independently derived) normative status that privileges the optimal model as an interesting and informative basis for exploration in ideal observer analysis or in the task analysis examples above. In the case of testimony, for example, there are infinitely many strategies one could conjure up and assess with respect to whether they could be feasibly implemented by human agents. Showing that an arbitrarily chosen one of them is not practically feasible would likely not feel like a major information gain in the way it does for the optimal strategy.

At the same time, the inherent flexibility of the human behavior, whereby we can typically choose from many possible strategies, means that even the space of potentially plausible descriptive

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theories is vast. The idea that the agent is trying to do something reasonable provides an important constraint, and it is likely no accident that those research areas within psychology that involve behaviors governed by the standard normative frameworks outlined in Part III are among those with the most detailed quantitative predictions. This is illustrated also by the development of neuroeconomics (Glimcher 2004, 2013) as motivated by the need for better organizing frameworks for cognitive neuroscience than traditional stimulus–response theories.

Finally, the rich inferential relationship between normative and descriptive helps us understand why the development of normative theory is pursued not only by philosophers but also by economists and psychologists: More detailed consideration of ways in which actual behavior deviates from a potential normative standard may also suggests ways in which that normative standard is incomplete (as in the case of deliberate ignorance mentioned above).

7. PART III: THE CLASSIC DOMAINS OF RATIONALITY RESEARCH

Beyond the ongoing development of the frameworks defining what it means to be rational and their methodological uses, the main focus of psychological research concerned with rationality has been the descriptive question of the extent to which human behavior lives up to rational standards.

This section provides an overview of key areas of research to that effect and, with that, a crude map of extant research. It will not, however, seek to provide comprehensive reviews of these areas, let alone definitive verdicts on human performance. The reasons for this are twofold: first, the sheer volume of literature; second, the fact that, in a general sense, the verdict on human behavior is effectively known, even without engaging with the details of empirical results. The world around us offers ample evidence of behavior that seems intuitively irrational, and the very question of the extent to which "is" and "ought" match requires a sense that they can and do diverge. At the same time, it seems hard to see how any normative standard that rendered all human behavior wholly irrational would ever be accepted as an appropriate standard in the first place. It would simply be too divergent from our actual intuitions and behavior to qualify as a candidate for what we ought to do.

Consequently, the answer to the question of whether human behavior matches up to the standards of rationality will clearly involve a glass half empty or half full, depending on one's perspective. That is, it will depend on how much the degree of deviation matters in any given context. Against that backdrop, it is unsurprising that assessments in the so-called rationality wars have deviated widely, but also that significant parts of those differences seem based more on rhetoric than on substantive disagreement concerning the actual level of performance (Samuels et al. 2002, Sturm 2012, Erickson et al. 2013, Gigerenzer 2025). This has been paralleled by criticisms of bias-focused research in social psychology (Krueger & Funder 2004). It thus seems doubtful that general verdicts on whether or not humans are rational are interesting or useful. Instead, it seems best left to specific interests and contexts to decide whether a given human performance level is practically consequential or theoretically informative for understanding psychological processes. So a broad outline of key topics seems most useful for a general reader.

7.1. Judgment

The main home of research on human rationality within psychology has been judgment and decision-making research (JDM). Here, work on judgment has concerned itself primarily with the extent to which human judgments of probability or risk conform to the normative strictures of probability theory. Initially, this work (see Peterson & Beach 1967 for a review) examined belief revision in light of new evidence. It used a paradigm where all relevant parameters for the application of Bayes' rule were explicitly defined by an experimental setup modeled on random draws from an urn containing colored balls of varying proportions (a standard model in

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probability theory). While the overarching assessment of performance tended to be positive, systematic deviations were found that implied underweighting of the diagnostic value of the evidence, leading to belief updates that were less extreme than normatively required, a phenomenon that came to be known as conservatism (Phillips & Edwards 1966, Edwards 1968). An extensive literature then sought to identify the precise cause of these misestimations (e.g., misperception of probabilities, mis-aggregation, anchoring on initial values, unwillingness to use the extreme ends of response scales, and so on) without ever reaching a conclusion (see Edwards 1968, Erev et al. 1994).

This was followed by Kahneman & Tversky's heuristics and biases program (Tversky & Kahneman 1974), which drew on text-based vignettes to identify a range of seeming biases such as base rate neglect (Kahneman & Tversky 1973), the conjunction fallacy (Tversky & Kahneman 1983), anchoring and adjustment (Tversky & Kahneman 1974), overconfidence (Tversky 1974), and misperceptions of randomness (Tversky & Kahneman 1973). Each of these turned into a topic of research in its own right. Each continues to attract new research to this day, which has surfaced a wealth of contributing factors to these seeming errors alongside alternative conceptions of these tasks that lead to more favorable interpretations of human performance (see, e.g., Tentori et al. 2013 on the conjunction fallacy and Hahn & Warren 2009 on perceptions of randomness). Alternative conceptions typically either point out deviations between experimenter-intended interpretations and participants' understanding of the task (e.g., Hilton 1995, Hertwig & Gigerenzer 1999) or offer modified normative standards (e.g., Pothos & Busemeyer 2022).

At the same time, there are sizeable bodies of research on judgment tasks that were initially motivated by concerns in other areas of psychology, such as health psychology and social psychology. Emblematic of this is the extensive literature on (seeming) unrealistic optimism (Weinstein 1980) mentioned above, which claimed to have found robust evidence (across hundreds of studies) that individuals seemingly underestimate their individual risk of experiencing adverse future events. This literature overlaps with research on confirmation bias and motivated reasoning that we return to below.

Finally, research on probability estimation has also been conducted in the context of Bayesian models of cognition (Griffiths & Tenenbaum 2006) and in the context of cognitive neuroscience (Yang & Shadlen 2007, Trommershauser et al. 2011), which has led, more recently, to a considerable broadening of the judgment tasks examined in the literature.

7.2. Decision-Making

Much of this overall historical trajectory is replicated in the other component of JDM research: decision-making. In fact, the early psychological research on examining potential deviations from EUT was conducted by Ward Edwards (for a review, see Edwards 1954), who also led the early work on judgment just described. Like that research, the early research on decision-making sought to evaluate performance in contexts where key parameters (probability and value) had explicit, objective specifications. The observed behavior was not readily brought in concordance with EUT on the assumption that participants were drawing on either those values or those objective probabilities. Kahneman & Tversky's heuristics and biases program was also as focused on decision-making as it was on judgment and gave rise to a variety of phenomena of interest (not only to psychologists but also to economists; Rabin 1998) and to a wave of alternative models seeking to be descriptively more accurate variants of EUT, including Kahneman & Tversky's (1979) own such model, prospect theory. Likewise, alternative construals of reasonableness (Simon 1957) prompted a search for alternative choice heuristics across a broad range of tasks (Gigerenzer et al. 1999). Consideration of rational agents as operating in particular environments also prompted a

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body of research concerned with potentially systematic differences between so-called decisions from description, where probability and utility information is explicitly provided, and decisions from experience, where probabilities must be learned (Hertwig et al. 2004; see Hertwig 2015 for a review). This latter question dovetails with the use of decision theory as an organizing framework for data in the context of cognitive neuroscience under the banner of neuroeconomics (e.g., Glimcher 2004, 2013; Camerer et al. 2005; Trepel et al. 2005). There, cognitive neuroscientists investigated tasks as diverse as motor movement (e.g., reaching and pointing; Trommershäuser et al. 2003, 2006), eye movements (e.g., Platt & Glimcher 1999, Navalpakkam et al. 2010, Schütz et al. 2012), joint decisions (Bahrami et al. 2010), or speed/accuracy trade-offs (e.g., Bogacz et al. 2006)—all framed (or reframed) in decision-theoretic terms. This also prompted consideration of the extent to which the same choices are better handled in different modalities or different parts of the system, such as perception versus cognition (Trommershäuser et al. 2008; but see also Jarvstad et al. 2013, 2014).

Moving away from paper and pencil evaluation of pairwise gambles (e.g., "Would you prefer option A, a 50/50 chance of winning 100 dollars, or option B, winning 45 dollars for sure?"), as was the standard tool of decision-making research in Kahneman & Tversky's heuristics and biases program, to a focus on rational agents conducting more naturalistic tasks within actual environments also makes clear that information search itself is a decision problem. This has long been known in the context of exploration versus exploitation trade-offs. Here, optimal policies can be difficult to discern (see, e.g., Cohen et al. 2007, Wu et al. 2018), and the consequences of exploration/exploitation trade-offs for decision-makers' samples may themselves provide alternative explanations of potential deviations from rational decision-making (see, e.g., Denrell & March 2001, Denrell 2007). However, even in experimental paradigms where they face only a single source, participants have been found to seemingly undersample evidence relative to normative expectations, collecting less evidence than they should (Hertwig & Pleskac 2008, 2010). This prompted extensive debate (including normative debate) about the diagnostic value of small samples (see Fiedler & Juslin 2006 for extensive discussion). At the same time, an increasing range of tasks, from category learning to games like 20 questions, have become the focus of research on information search (e.g., Nelson 2005; Crupi et al. 2018; Meder et al. 2019, 2022); this literature overlaps with research concerns that have developed in the context of the study of reasoning (in the context of Wason's selection task) discussed below. Finally, there has been long-standing interest in potential inconsistencies in other aspects of choice, such as the way we factor in time, that is, temporal discounting (for a review, see Camerer et al. 2005, Grüne-Yanoff 2015).

Beyond EUT, there has also been extensive experimental research on the concordance of human behavior with the rational prescriptions of game theory. Canonical games such as the Prisoner's Dilemma (in single-shot versions and in iterated versions; on the normative implications of that distinction, see Aumann 1959, Axelrod 1980) or the Ultimatum Game have prompted hundreds of studies (see, e.g., Thaler 1988, Camerer & Thaler 1995, Kollock 1998), including within cognitive neuroscience (e.g., Sanfey et al. 2003; see Camerer et al. 2005).

7.3. Reasoning

Historically remarkably separate from JDM research was rationality-focused research on reasoning. The point of departure for the latter was a long-standing philosophical intuition that equated rationality with logic and took logic, in effect, to be constitutive of thought (for references, see, e.g., Steinberger 2022). This prompted experimental investigation into the way in which logical reasoning is achieved (Braine 1978, Rips 1983, Johnson-Laird et al. 1992). Significant differences between normative responding and observed behavior on what were ostensibly logical reasoning

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tasks soon became apparent [Evans 2002, 2013 (1982)]. These span syllogistic reasoning, transitive reasoning, and various forms of logically valid inferences in propositional logic such as reasoning with disjunctions or reasoning with conditionals.

Of these, reasoning with conditionals (broadly, if-then statements, though their linguistic form may vary; see DeClerck & Reed 2012) proved pivotal. It focused on classic logical fallacies—denying the antecedent and affirming the consequent—and lay reasoners' ability to distinguish these logically invalid inferences from the logically valid inferences modus ponens and modus tollens. This provided reach into research on argument and argumentation (to be discussed in a moment). Through Wason's classic (1968) selection task, research on conditionals also linked to falsificationism as a fundamental strategy for science (Popper 1963; but see Howson & Urbach 1989), research on confirmation bias (Wason 1960, Nickerson 1998), and, ultimately, research on information search.

Oaksford & Chater's (1994) seminal paper argued that participant responses on Wason's selection task were, in fact, rational if one assumed probability theory instead of classical logic as the normative lens, and that participants' choices, under certain assumptions, maximized expected information gain. This argument echoed earlier probabilistic assessments of seemingly confirmatory strategies in hypothesis testing (e.g., Klayman & Ha 1987). It also affirmed not only the general point that experimenter and participant interpretations of behavioral tasks may diverge (Hilton 1995) but also that establishing an appropriate mapping between a formalism and a task may be fraught with difficulty. More specifically, the natural language meaning of the conditional is not well captured by classical logic, which itself parallels difficulties in both the judgment and the reasoning literature surrounding the meaning of natural language "and" (e.g., Dulany & Hilton 1991, Hertwig et al.2008).

Debate on the meaning of conditionals continues (e.g., Evans et al. 2003, Over & Evans 2003, Douven & Verbrugge 2013, Douven et al. 2018, Krzyżanowska et al. 2021, López-Astorga et al. 2022), so that questions about this important aspect of reasoning and rationality remain unresolved.

Although debate about the appropriateness of Oaksford & Chater's reanalysis of selection task and other seemingly deductive tasks in probabilistic terms has continued, this has not hindered what has been labeled the probabilistic turn in the psychology of reasoning (Over 2009, Oaksford & Chater 2020).

This shift has brought reasoning research into closer contact with JDM research. It has also dovetailed with new lines of research examining probabilistic reasoning in ways that go beyond the comparatively simple judgments of traditional JDM research by examining performance on evidential reasoning involving multivariable interactions (e.g., Pilditch et al. 2019, 2021), as found in everything from everyday reasoning through scientific reasoning to legal reasoning (Lagnado 2021). By virtue of a set of common underlying formal tools, this includes aspects of causal reasoning (Meder et al. 2014, Sloman & Lagnado 2015)—for example, diagnostic reasoning from evidence to putative causes (e.g., Meder & Mayrhofer 2017). Such reasoning may be challenging particularly where multiple causes compete, as in the case of explaining away, whereby evidence for one cause may lower the probability of another, alternative cause (Rottman & Hastie 2014, Tešić et al. 2020).

At the same time, probability theory has been applied to what were previously largely intuitive notions of bias and irrationality, such as motivated reasoning (Kunda 1990, Hahn & Harris 2014) and biased assimilation of evidence (Lord et al. 1979, Jern et al. 2014), leading to more nuanced appraisals.

In short, the use of a shared normative framework has allowed greater integration between previously fragmented and isolated aspects of behavioral research concerned with rationality.

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7.4. Argument

This integration extends into a final domain that touches on rationality: argumentation. Argumentation researchers distinguish many types of argument such as quarrels, disputes, and more; but much of the focus of argumentation research, both theoretically and empirically, has been on rational debate and rational argument (van Eemeren 2001; for a review, see Hahn & Oaksford 2012). Some of the concerns of this literature overlap with those of research on critical thinking and with empirical studies concerned with improving reasoning and argument skills (e.g., Kuhn & Crowell 2011. El Soufi & See 2019). Fallacies of argument are one such point of overlap (e.g., Rips 2002, Neuman 2003, Weinstock et al. 2006).

The so-called logical fallacies of denying the antecedent and affirming the consequent have already been mentioned, but beyond these there is a much wider catalogue of argument forms that may trick unwary reasoners (dating in origin all the way back to Aristotle), such as ad hominem arguments (Bhatia & Oaksford 2015), arguments from ignorance (Hsu et al. 2017), circular arguments (Rips 2002), slippery slope arguments (Corner et al. 2011), and more (Hahn & Oaksford 2007). These fallacies not only are of interest in their own right but also—together with probabilistic and logical errors—have featured as possible hallmarks and indicators of conspiracy theory (for a review, see Gagliardi 2025) and science denialism (e.g., Cook et al. 2018).

As noted, Oaksford & Chater's work made clear that the logical fallacies of denying the antecedent and affirming the consequent—viewed as probabilistic, uncertain arguments—are not fallacious (though misperception lives on in Wikipedia pages and Internet discourse). Similarly, argumentation scholars had come to realize that many of these other fallacies have instantiations or interpretations that, intuitively, seem acceptable arguments (Finocchiaro 1981, Woods & Walton 2019). The normative framework of probability theory has helped explicate and clarify these intuitions (see Hahn 2020). This, in turn, has opened up both new theoretical avenues and new avenues for empirically evaluating the rationality of intuitions about everyday arguments (Corner & Hahn 2009, Hahn 2020).

Needless to say, the task of understanding what argument is being made by a speaker far exceeds the difficulties of understanding the precise meaning of connectives such as "and" or natural language conditionals. Understanding what argument is being made involves understanding the speaker's intended meaning, that is, what in the study of language is referred to as pragmatics. This involves not only disambiguating the meaning of what is said but also considering what is implied. For this, rational standards themselves provide interpretative tools in that we resolve ambiguity with the principle of charity (e.g., Govier 1987; see also Oaksford 2014). Given multiple interpretations of what someone is saying, we may pick as our default the interpretation that renders what they are saying most sensible (though further evidence may, of course, force us to abandon that interpretation). The context of argument thus seems a good place to mention also that the formal tools of rationality research play a role in explicating how meanings are derived (e.g., Goodman & Frank 2016).

8. PART IV: INDIVIDUALS AND COLLECTIVES

One strand of research on rationality that is emerging as a likely focal point of increasing interest concerns the relationship between the rationality of individuals and the rationality of collectives. There has long been interest in the relationship between putatively rational individual and collective outcomes inasmuch as one of the most prominent uses of rational actor models has been within economics, where there is a long tradition of using them to supply the micro behaviors with which to build macroeconomic theories. Historically, it was precisely critique of these rational actor models as descriptively valid (with attendant implications for macroeconomic theorizing) that

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motivated much of the most influential work on human rationality (Simon 1957, Kahneman & Tversky 1979) and continues to do so within behavioral economics to this day. Furthermore, we saw above in the context of game theory that individual and collective rationality may readily come apart. In the context of decisions, this is also apparent from Arrow's (1950) impossibility theorem. Arrow's theorem demonstrates that rational preferences of individuals may fail to aggregate into rational preferences of the collective comprised of these individuals given nondictatorial aggregation, for example, by majority vote. However, it has also been shown that Arrow's result can be applied to certain contexts of belief aggregation (List & Pettit 2004).

There are consequently further questions to be asked about individuals and collectives. For one, it is of interest to what extent group deliberation and information exchange improve individuals' performance, enhancing their rationality (see, e.g., Rahwan et al. 2014, Mercier et al. 2015, Mercier & Claidière 2022). There are also questions to be asked about the epistemic rationality of collectives themselves (see, e.g., Hahn 2022), in keeping with a renewed wider interest in wisdom of the crowd phenomena and collective intelligence. Both of these endeavors fit also into a wider shift from individual-focused research to attempts to take into account the fundamentally social nature of human beings (Dingemanse et al. 2023). Finally, the relationship between the beliefs of individuals and of collectives is of interest. Mayo-Wilson et al.'s (2011) independence thesis asks to what extent individual and collective rationality are independent, such that a group of rational individuals may give rise to an irrational collective, or, conversely, a rational collective may derive from irrational individuals. We know already from List & Pettit's (2004) work that such dissociation is possible. We do not, however, currently understand well enough how likely such dissociations are or under what circumstances they arise. This will require theoretical work of which the analyses aimed at phenomena typically taken to be indicative of the irrationality of the individual actors, like herding (Banerjee 1992) or polarization (e.g., Hahn & Harris 2014, Jern et al. 2014, Dorst 2023), are one example. It will also require simulations that probe the conditions that make dissociation likely (e.g., Mayo-Wilson et al. 2013), and it will require empirical research to examine when these circumstances actually occur.

Finally, there may be aspects of rationality itself that are better conceived of as being in some way social (for example, as pertaining to social expressions of beliefs and commitments) rather than individualistic (see, e.g., Oaksford 2025).

9. PART V: RATIONALITY IN THE REAL WORLD

This review has covered over a hundred studies. In the sprawling research literature, there are thousands upon thousands more. This reflects the central role of human rationality to our understanding of ourselves and to a myriad of facets of human behavior. Correspondingly, there can be few topics, if any, for which psychological research impacts so many other disciplines. However, the same questions concerning the extent to which psychological research meaningfully elucidates actual real-world behavior that arise all across psychology are also found here. If anything, for rationality research these questions are even more acute.

The inference from controlled behavioral experiment to real-world behavior is problematic enough even when we are trying to understand behavior as is. Rationality research, in addition, compares behavior to standards whose realism can itself be questioned. This is further exacerbated by the generality of the normative standards themselves: As discussed at the outset, by nature, the standards that our normative frameworks provide are abstract. As a consequence, they apply to concrete problems that will differ in a myriad of relevant ways. Chief among these is scale: Motor movements, the purchase of lottery tickets, pandemic responding, or averting climate disaster could not be any more different. Yet the same constraints are meant to apply.

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In fact, a considerable chunk of the development of our current normative frameworks was motivated by the looming existential threats of the Cold War (Erickson et al. 2013). So it is not that this breadth of applicability has not been explicitly considered. These very different contexts do, however, give different emphases to the wider relationship between rationality and reason. From a purely descriptive perspective, it seems implausible that the underlying behavioral processes across such divergent contexts are meaningfully the same.

This translates directly into questions about what the appropriate domain of generalization is intended to be when an experimental study claims evidence that human decisions, for example, are biased in a particular way. Yet this is not a question that studies tend to address explicitly, and that gap will loom large when behavioral scientists are called to advise on real-world decisions.

Experimental control is hard everywhere in psychology, but it becomes ever harder the more unconstrained the means are by which a task can be achieved (relatedly, see Lejarraga & Hertwig 2021). Rationality research has skirted that issue with dual systems/dual process theories (Evans 2017). Aside from the intrinsic problems of such theories, packaging into a single "analytic route" decisions about everything—from hypothetical, paper-and-pencil math problems in classic behavioral experiments to real-world decisions about jobs, homes, or partners—creates a putative psychological process with more semblance of unity than it could possibly deliver.

Robust experimental demonstrations of irrationality need a clearly articulated null hypothesis of rational behavior: If a rational agent can generate that same behavior, the empirical test is flawed. In addition, participant interpretations need to match the experimenter's view of the task. These standards are hard, and this review has highlighted multiple failures to meet them (e.g., unrealistic optimism research and conditional reasoning, to name but two). Yet even empirical studies that meet these standards face yet another problem: that the incentives and consequences at play simply do not match real-world circumstances. No experimental study will be able to manipulate participants' exposure to significant risk. That, in itself, limits transfer to important real-world situations.

We do, however, have important observations on irrationality in highly consequential circumstances, like wars or pandemics, from historical records. It is regrettable that psychology does not engage more fully with these. There is precedent, for example, in cross-disciplinary collaboration on deliberate ignorance (Hertwig & Ellerbrock 2022). It would be beneficial to rationality research to see more of this.

10. PART VI: OUTLOOK

This review has sought to illustrate the particular nature of psychological research concerned with rationality and how it is distinct from other areas of psychological inquiry. It has sought to make clear why there is so much interest and emphasis in this area on discerning levels of competence as well as how and why this is surprisingly hard. It has sought to illustrate also how normative theories provide functional explanations, provide insights into tasks and the suitability of methods, and, in multiple ways, are employed as a methodological tool.

These different facets conspire to give psychological research on rationality (broadly construed) much less of a focus on psychological process than is familiar from other areas of psychology. They mean also that research on rationality is far richer, and about far more, than providing answer to the question of how rational we are. That question already has an obvious answer inasmuch as humans are neither fully rational nor completely irrational but rather are somewhere in between. It is the details that are interesting and consequential. And that detailed picture is still far from complete.

The utility of rationality research, arguably, does not lie in seeking some overall verdict. Crucially, our normative frameworks are still developing, as is our recognition of the domains to which

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we can apply these tools. Comparing the current review back to the excellent survey by Shafir & LeBoeuf (2002), the last *Annual Review of Psychology* article on rationality, makes clear how much those underlying drivers have broadened the field; and there is clearly more yet to come.

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